

Cell & Molecular
Biology Lab Activity:
Characterization of a Plastid Genome

Name: BATIGULAO, JEHIAH BLESS T.

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I. Introduction

The plastid genome, or plastome, is the genetic material found within plant plastids. In photosynthetic plants, the chloroplast genome contains genes involved in photosynthesis, transcription, translation, and other essential plastid functions. Most flowering-plant plastomes have a characteristic quadripartite structure consisting of a large single-copy (LSC) region, a small single-copy (SSC) region, and two inverted repeats (IRs).

For this activity, the complete chloroplast genome of *Allium sativum* L. was selected from the NCBI Nucleotide database. The genome was analyzed using Galaxy to obtain sequence statistics, while the annotated GenBank record was examined to characterize its genes and other genomic features.

II. Data Source

Genus	<i>Allium</i>
Species	<i>Allium sativum</i> L.
Common name	Garlic
Family	Amaryllidaceae
Genome	Chloroplast genome
NCBI accession	NC_031829.1
Genome size	153,172 bp
Topology	Circular
Database	NCBI Nucleotide/RefSeq
Analysis platform	Galaxy

III. Methodology

1. Selection of Plant Genus

The genus *Allium* was selected after checking the class spreadsheet to ensure that it had not already been assigned to another student. The NCBI Nucleotide database was then searched to confirm the availability of a complete plastid genome within the genus.

2. Genome Selection and Download

A complete chloroplast genome of *Allium sativum* L. was selected using NCBI accession NC_031829.1. The genome was downloaded in FASTA format for sequence analysis. The annotated GenBank record was also retained for examination of genomic features.

3. Galaxy Analysis

The FASTA sequence was uploaded to a personal Galaxy account. A FASTA statistics tool was used to determine the genome length, number of sequences, GC content, nucleotide composition, N50/L50 values, and gaps.

4. Genome Annotation Analysis

The annotated GenBank record was examined to identify protein-coding genes, tRNA genes, rRNA genes, pseudogenes, intron-containing genes, and duplicated genes associated with the inverted-repeat regions.

IV. Results

Genome Identification and General Characteristics

The selected species for this study was *Allium sativum* L., commonly known as garlic, belonging to the family Amaryllidaceae. The complete chloroplast genome was obtained from the NCBI Nucleotide database under accession NC_031829.1. The genome is reported as a circular molecule with a total length of 153,172 bp.

Galaxy-based FASTA statistics confirmed that the downloaded sequence consisted of one sequence record with a total length of 153,172 bp. The overall GC content was 36.68%, with no ambiguous bases (N = 0) and no detected gaps. The scaffold and contig N50 values were both 153,172 bp, while the corresponding L50 values were 1, indicating that the complete plastome was represented by a single continuous sequence.

Plastome Organization

The *A. sativum* chloroplast genome exhibits the typical quadripartite organization commonly observed in angiosperm plastomes. It consists of a Large Single-Copy (LSC) region of 82,035 bp, a Small Single-Copy (SSC) region of 18,015 bp, and two Inverted Repeat (IR) regions of 26,561 bp each. These regions together account for the complete genome length of 153,172 bp.

The presence of two inverted repeat regions contributes to the characteristic structural organization of the plastome and results in the arrangement LSC–IR–SSC–IR.

Annotated Gene Content

Based on the annotation of the GenBank record, the chloroplast genome contains 135 annotated gene features, including 83 coding sequence (CDS) features, 38 tRNA genes, and 8 rRNA genes. The annotation also includes 6 pseudogene features.

The protein-coding genes represent different functional categories associated with photosynthesis, transcription, translation, and other essential plastid processes. Examples include *rbcL*, which encodes the large subunit of RuBisCO and participates in carbon fixation; *psaA* and *psaB*, which encode core components of photosystem I; and *atpA* and *atpB*, which are involved in ATP synthase function. Genes such as *rpoA*, *rpoB*, and *rpoC* contribute to transcription, while ribosomal protein genes support protein synthesis within the chloroplast.

RNA Genes and RNA-Processing Features

The annotation includes 38 tRNA genes and 8 rRNA genes, reflecting the capacity of the plastid genome to encode components required for its own gene-expression machinery. Several

annotated genes also contain introns. These intron-containing genes are relevant to plastid RNA processing because their transcripts require intron removal before producing functional RNA or protein products.

Pseudogenes and Other Notable Features

Six pseudogene features were identified in the GenBank annotation, including annotations associated with *rps2*, *rps16*, *infA*, *ycf15*, and *ycf1*. Pseudogene annotations may represent gene remnants or sequences that have undergone changes affecting their ability to encode functional products.

The presence of duplicated regions is also an important characteristic of the plastome because the two IR regions contain homologous sequences. Overall, the genome maintains the typical structural organization of angiosperm chloroplast genomes.

4.6 GC Content and Sequence Characteristics

The overall GC content determined through Galaxy was 36.68%. The sequence contained 47,983 adenine (A), 49,010 thymine (T), 28,606 cytosine (C), and 27,573 guanine (G) bases. No ambiguous bases or gaps were detected. These results indicate that the downloaded sequence represents a complete and continuous plastid genome suitable for downstream analysis.

Comparison of Plastid and Mitochondrial Genomes

Plastid and mitochondrial genomes are both organellar genomes and share several characteristics, including their presence within eukaryotic cells, predominantly non-nuclear inheritance, and their usefulness in evolutionary and phylogenetic studies. However, they differ substantially in their biological functions, genome organization, size, gene content, and patterns of structural variation.

Characteristic	Plastid Genome	Mitochondrial Genome
Cellular location	Chloroplast/plastid	Mitochondrion
Main function	Photosynthesis and plastid-related metabolism	Cellular respiration and energy production
Organization	Usually circular and quadripartite in plants	Often circular but structurally more variable
Typical size	Generally tens to hundreds of kb	Usually larger and more variable in plants
Gene content	Photosynthetic, transcriptional, translational genes	Primarily respiratory and mitochondrial gene-expression genes
Inheritance	Commonly maternal in plants	Commonly maternal in plants
Recombination	Generally limited	More extensive structural recombination may occur
Research applications	Plant phylogeny, DNA barcoding, population studies	Maternal lineage, mitochondrial evolution, plant cytoplasmic genetics

Practical Applications

Plastid genomes are particularly useful for studying plant evolutionary relationships because they are relatively conserved, occur in multiple copies per cell, and contain regions that can provide phylogenetic information at different taxonomic levels. They are also useful for species identification, comparative genomics, and studies of chloroplast evolution.

Mitochondrial genomes provide complementary information, particularly for studies involving mitochondrial inheritance, cellular respiration, and mitochondrial genome evolution. Using plastid and mitochondrial data together with nuclear genomic information can provide a more comprehensive understanding of plant evolutionary history and genetic relationships.

Overall Interpretation

The analysis confirmed that *Allium sativum* possesses a complete chloroplast genome of 153,172 bp with a GC content of 36.68%. The genome displays the characteristic quadripartite structure of angiosperm plastomes and contains diverse genes associated with photosynthesis, transcription, translation, and plastid gene expression. The combination of NCBI annotation and Galaxy sequence statistics provided consistent evidence that the downloaded sequence represents a complete and continuous chloroplast genome.

Github Repository: https://github.com/uzei23/cmb_plastid-genome_Allium_BATIGULAO/tree/main

Galaxy Repository: <https://usegalaxy.org/u/jehiah/h/plastid-allium-batigulao>